

Electrical Circuits I

ENGR 2405

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Chapter 02 – Basic Laws

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Overview

- Resistors
- Ohm's Law
- Circuit topology, KVL, KCL

Resistivity

- Materials tend to resist the flow of electricity through them - this property is called “resistance”
- The resistance of an object is a function of its length, l , and cross sectional area, A , and the material’s resistivity:

$$R = \rho \frac{l}{A}$$

Resistivity of Common Materials

Resistivities of common materials.

Material	Resistivity ($\Omega\cdot\text{m}$)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^2	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator

Ohm's Law

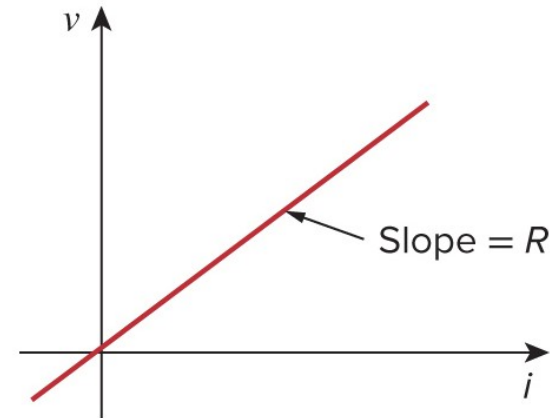
- In a resistor, the voltage across it is directly proportional to the current flowing through it or,
 $V = IR$
- The resistance of an element is measured in units of Ohms, Ω (V/A)
- The higher the resistance, the less current will flow through for a given voltage
- NOTE: Ohm's law requires conforming to the passive sign convention

Short and Open Circuits

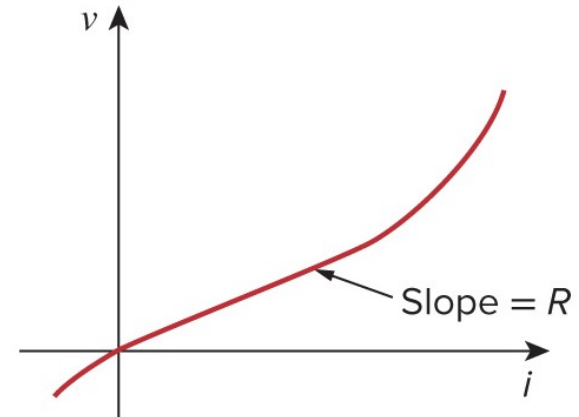
- A connection with almost zero resistance is called a short circuit
 - Ideally, any current may flow through the short
 - In practice this is a connecting wire
- A connection with infinite resistance is called an open circuit
 - Here no matter the voltage, no current flows

Linearity

- Not all materials obey Ohm's Law
- Resistors that do are called "linear" resistors because their current voltage relationship is always linearly proportional
- Diodes and light bulbs are examples of non-linear elements



(a)



(b)

Power Dissipation

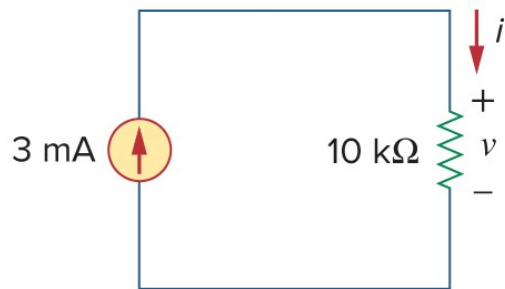
- Running current through a resistor dissipates power

$$P = IV = I^2 R = \frac{V^2}{R}$$

- NOTE: the power dissipated is a non-linear function of current or voltage
- Power dissipated is always positive
- A resistor can never generate power

Practice Problems

- 2.1 The essential component of a toaster is an electrical element (a resistor) that converts electrical energy to heat energy. How much current is drawn by a toaster with resistance of 15Ω at 110V ?
- 2.2 What is V , G , and P for the following circuit?



- 2.3 A resistor absorbs an instantaneous power of $30 \cos^2 t$ mW when connected to a voltage source $v = 15 \cos t$ V. Find i and R .

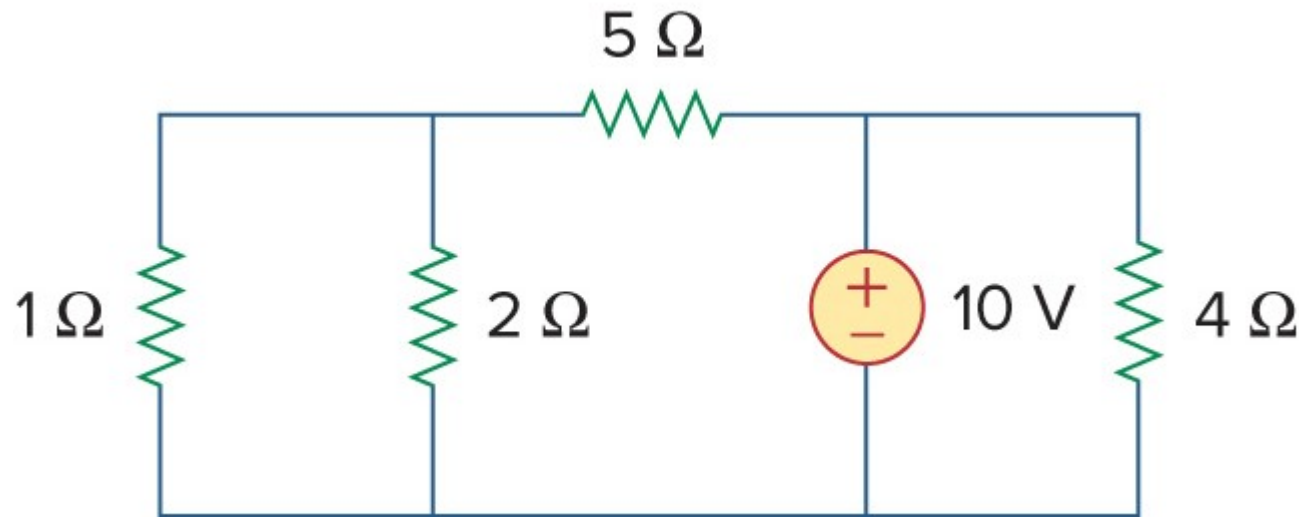
Nodes Branches and Loops

- Circuit elements can be interconnected in multiple ways
- We can use network topology concepts to help understand these options
 - A branch represents a single element such as a voltage source or a resistor
 - A node is the point of connection between two or more branches
 - A loop is any closed path in a circuit

Network Topology

- A loop is independent if it contains at least one branch not shared by any other independent loops
- Two or more elements are in series if they share a single node and thus carry the same current
- Two or more elements are in parallel if they are connected to the same two nodes and thus have the same voltage

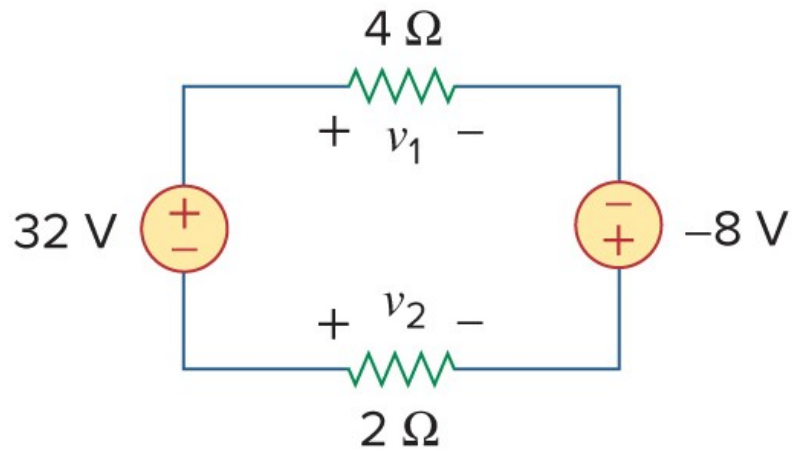
Practice Problems



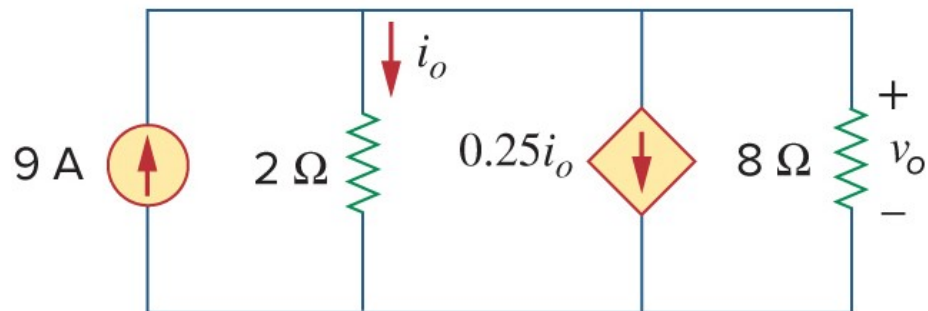
2.4 How many branches and nodes are in this circuit? Which elements are in parallel and which ones are in series?

Practice Problems

2.5 Find v_1 and v_2



2.7 Find v_o and i_o



Kirchoff's Laws

Ohm's law is not sufficient for circuit analysis, thus we also need Kirchoff's two laws complete the needed tools:

- Kirchoff's Current Law
- Kirchoff's Voltage Law

Kirchoff's Current Law (KCL)

- KCL is based on conservation of charge
- It states that the algebraic sum of currents entering a node (or a closed boundary) is zero, or:

$$\sum_{n=1}^N i_n = 0$$

Kirchoff's Voltage Law (KVL)

- KVL is based on conservation of energy
- It states that the algebraic sum of voltages around a closed path (or loop) is zero, or:

$$\sum_{m=1}^M v_m = 0$$

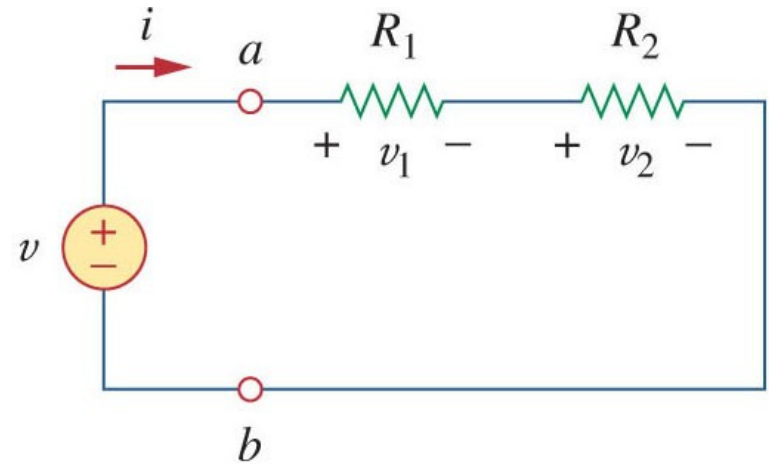
Series Resistors

- Two resistors are considered in series if the same current pass through them
- Take the circuit shown:
 - Applying Ohm's law to both resistors:

$$v_1 = iR_1 \quad v_2 = iR_2$$

- If we apply KVL to the loop we have:

$$-v + v_1 + v_2 = 0$$



Series Resistors

- Combining the two equations:

$$v = v_1 + v_2 = i(R_1 + R_2) = iR_{eq}$$

- From this we can see there is an equivalent resistance of the two resistors:

$$R_{eq} = R_1 + R_2$$

- For N resistors in series:

$$R_{eq} = \sum_{n=1}^N R_n$$

Voltage Division

- The voltage "drop" across any one resistor can be known
- The current through all the resistors is the same, so using Ohm's law:

$$v_1 = \frac{R_1}{R_1 + R_2} v \quad v_2 = \frac{R_2}{R_1 + R_2} v$$

Parallel Resistors

- When resistors are in parallel, the voltage drop across them is the same:

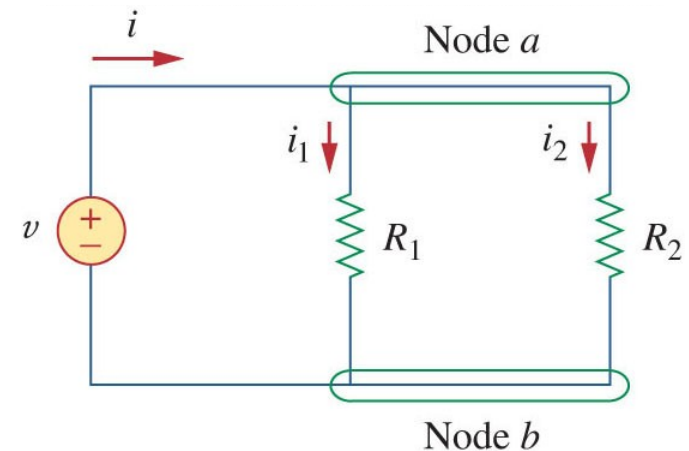
$$v = i_1 R_1 = i_2 R_2$$

- By KCL, the current at node a is

$$i = i_1 + i_2$$

- The equivalent resistance is:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$



Current Division

- Given the current entering the node, the voltage drop across the equivalent resistance will be the same as that for the individual resistors:

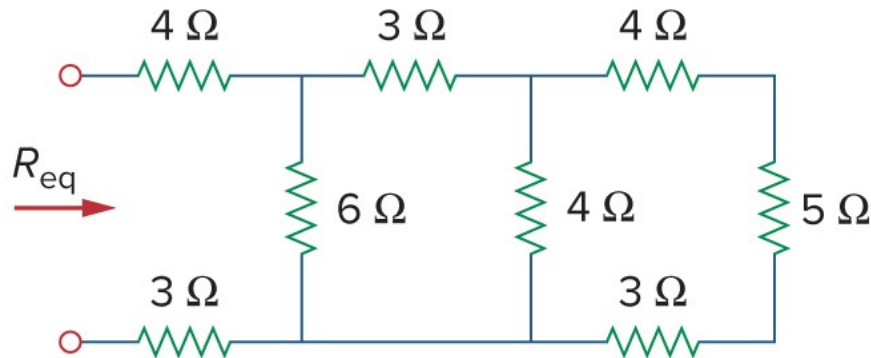
$$v = iR_{eq} = i \frac{R_1 R_2}{R_1 + R_2}$$

- This can be used in combination with Ohm's law to get the current through each resistor:

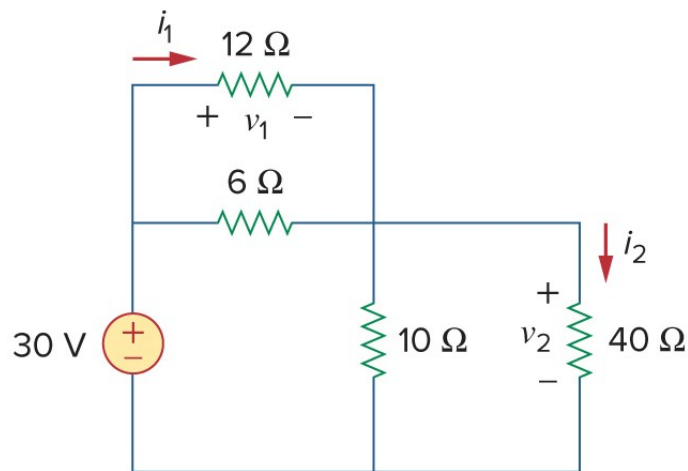
$$i_1 = i \frac{R_2}{R_1 + R_2} \quad i_2 = i \frac{R_1}{R_1 + R_2}$$

Practice Problems

2.9 Find R_{eq}

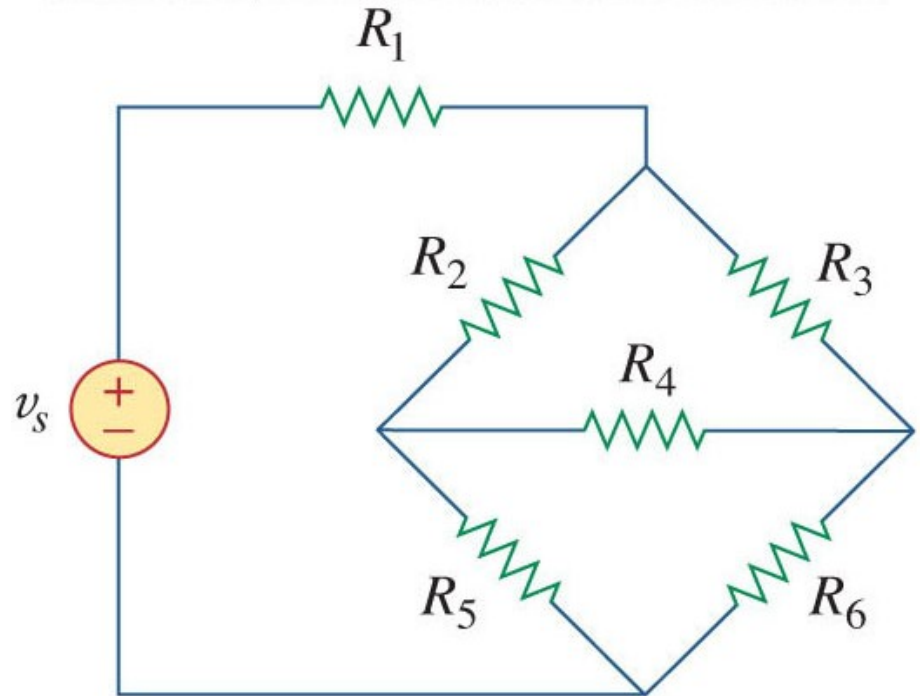


2.12 Find v_1 , v_2 , i_1 , i_2 , $P(12\ \Omega)$, and $P(40\ \Omega)$



Wye-Delta Transformations

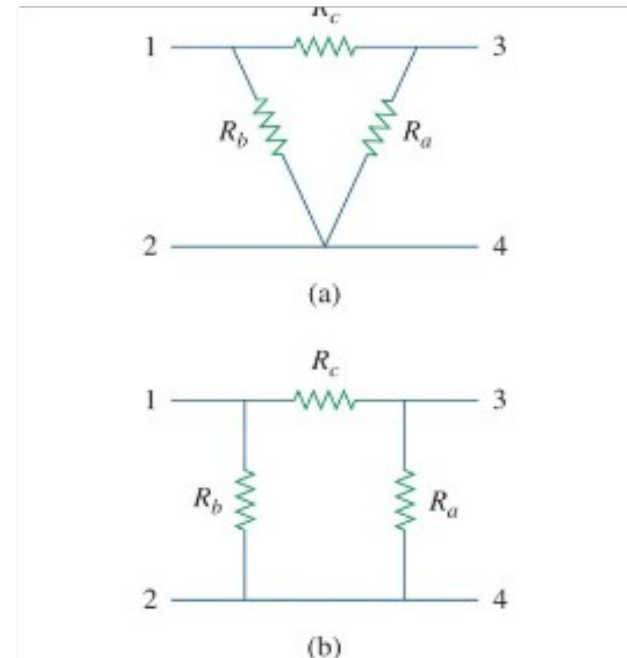
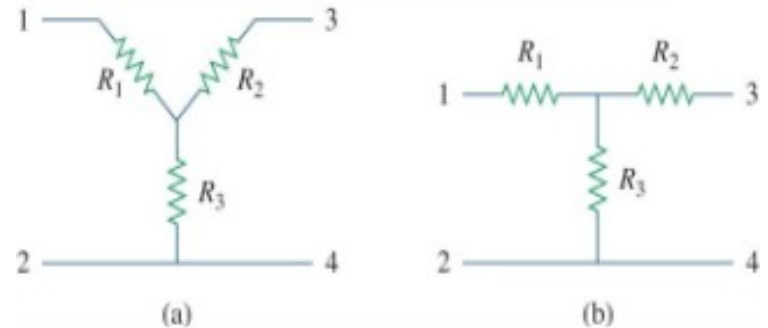
- There are cases where resistors are neither parallel nor series
- Consider the bridge circuit shown here
- This circuit can be simplified to a three-terminal equivalent



Wye-Delta Transformations II

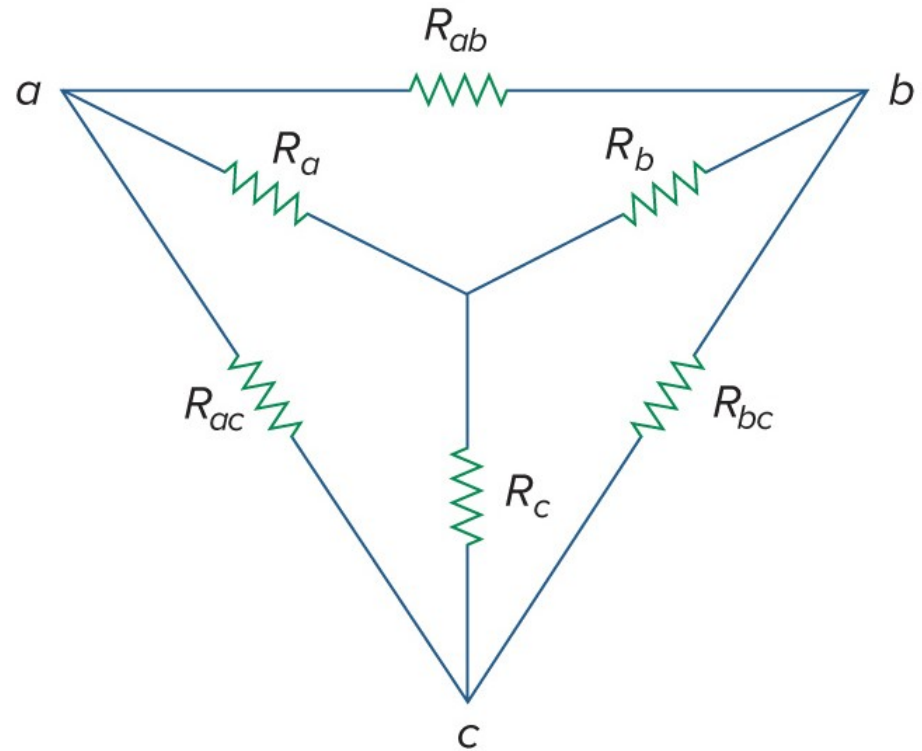
Two topologies can be interchanged:

- Wye (Y) or tee (T) networks
- Delta (Δ) or pi (Π) networks.
- Transforming between these two topologies often makes the solution of a circuit easier



Wye-Delta Transformations III

- The superimposed wye and delta circuits shown here will be used for reference.
- The delta consists of the outer resistors, labeled ab , bc , and ac .
- The wye network are the inside resistors, labeled a , b , and c .



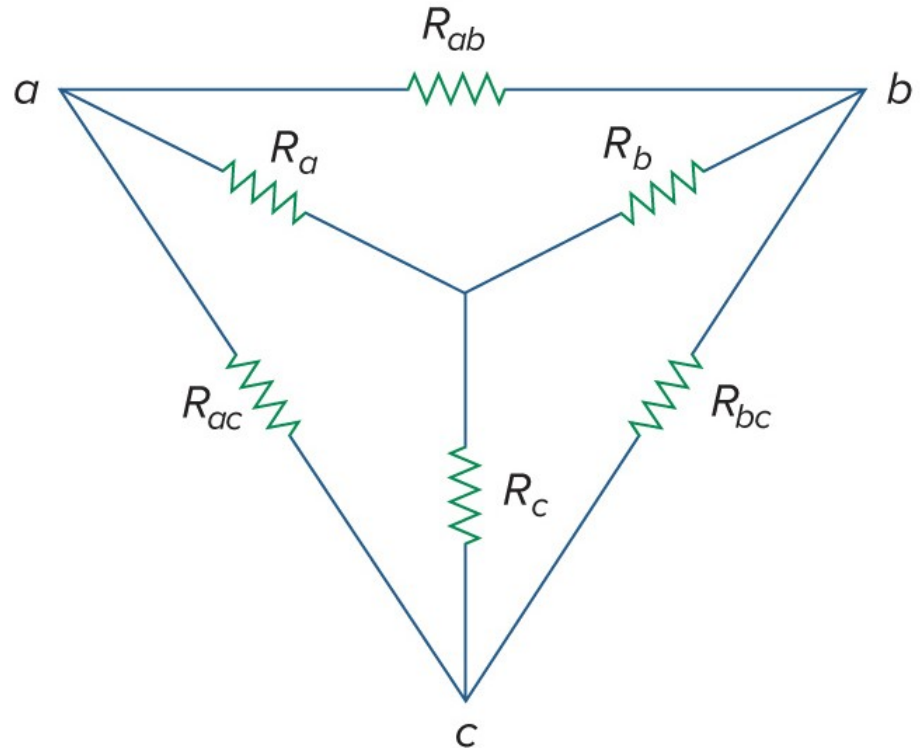
Delta to Wye

- The conversion formula for a delta to wye transformation are:

$$R_a = \frac{R_{ab} R_{ac}}{R_{ab} + R_{bc} + R_{ac}}$$

$$R_b = \frac{R_{ab} R_{bc}}{R_{ab} + R_{bc} + R_{ac}}$$

$$R_c = \frac{R_{ac} R_{bc}}{R_{ab} + R_{bc} + R_{ac}}$$



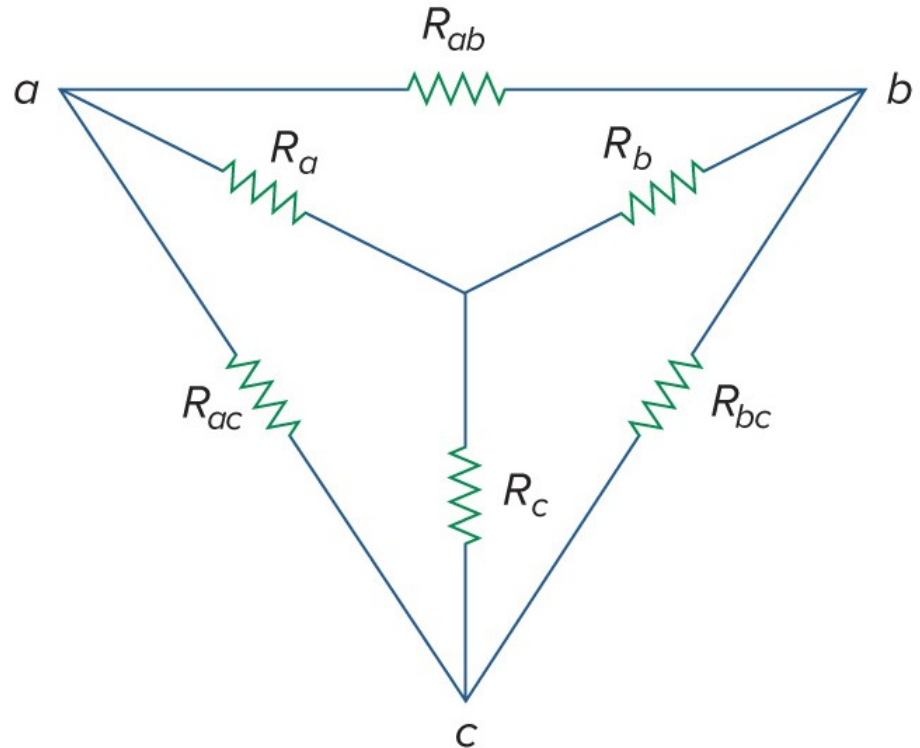
Wye to Delta

- The conversion formula for a wye to delta transformation are:

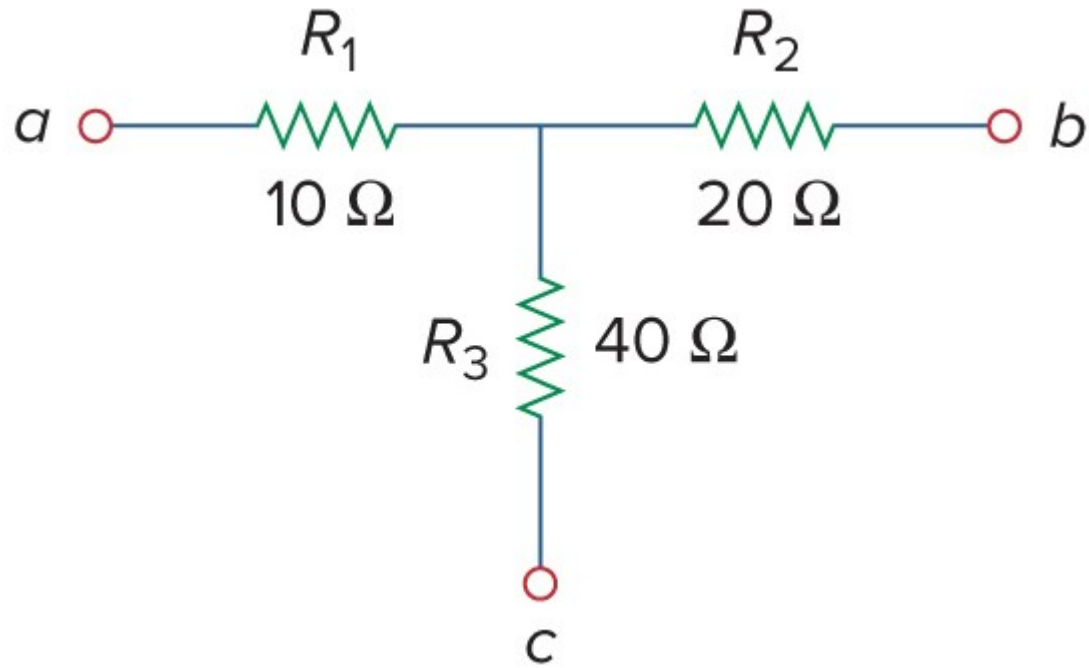
$$R_{ab} = \frac{R_a R_b + R_b R_c + R_a R_c}{R_c}$$

$$R_{bc} = \frac{R_a R_b + R_b R_c + R_a R_c}{R_a}$$

$$R_{ac} = \frac{R_a R_b + R_b R_c + R_a R_c}{R_b}$$



Practice Problems



2.14 Transform from wye to delta